

OpenCTI Deployment Report

My task is

The goal of this task was to deploy OpenCTI from scratch and prepare it for integration with external threat intelligence to platform AlienVault OTX.

Initial Environment

- Operating System: Windows
 - I didn't have pre-installed something (Docker, Git, etc. not installed)
-

Tools Used

- Docker Desktop (for containerized deployment)
 - Git (to download OpenCTI source)
 - Web Browser (for accessing that through link <http://localhost:8080/dashboard>)
-

Deployment Steps

Installed Docker Desktop

- Downloaded Docker Desktop for Windows
- Enabled WSL2 and virtualization
- Resolved initial issue:

"Virtualization support not detected"

First I opened Virtualization in BIOS to check if it was enabled and it was actually enabled, but then I opened Docker again and saw that I need to update ws from command by typing command "ws -update"

After that it worked!

Installed Git

- Installed Git to enable repository cloning
- Verified installation using:

- `git --version` (in cmd)

Downloaded OpenCTI

`git clone https://github.com/OpenCTI-Platform/docker.git` - To download OpenCTI from Github

`cd docker` – to move already to created folder and to make changes in notepad for having an access to OpenCTI

Configured Environment File

`copy .env.sample .env`

texted **notepad .env**

Modified key parameters:

`OPENCTI_ADMIN_EMAIL=admin@local.com`

`OPENCTI_ADMIN_PASSWORD=Admin12345!`

`OPENCTI_ADMIN_TOKEN=<UUID>`

`OPENCTI_HEALTHCHECK_ACCESS_KEY=<UUID>`

`OPENCTI_ENCRYPTION_KEY=<Generated Secret>`

Started OpenCTI

`docker-compose up -d` (To start it for work)

Issues Encountered & Resolutions

Issue 1: Docker virtualization error

- Error: Virtualization not detected
- Solution: Enabled virtualization in BIOS

Issue 2: Git not recognized

- Cause: Git not installed

- Solution: Installed Git
-

Issue 3: OpenCTI container unhealthy

- Error: "Token must be a valid UUID"
- Cause: Default token value used (ChangeMe_UUIDv4) MOST PROBABLY I JUST FORGOT TO SAVE AND THAT IS WHY I JUST DELETED NOTEPAD, CREATED A NEW ONE AND SAVED MY NEW CREDENTIALS.
- Solution:
 - Generated valid UUIDs
 - Updated .env
 - Recreated containers using:
 - docker-compose down -v TO DELETE EVERYTHING

docker-compose up -d RECREATE AGAIN

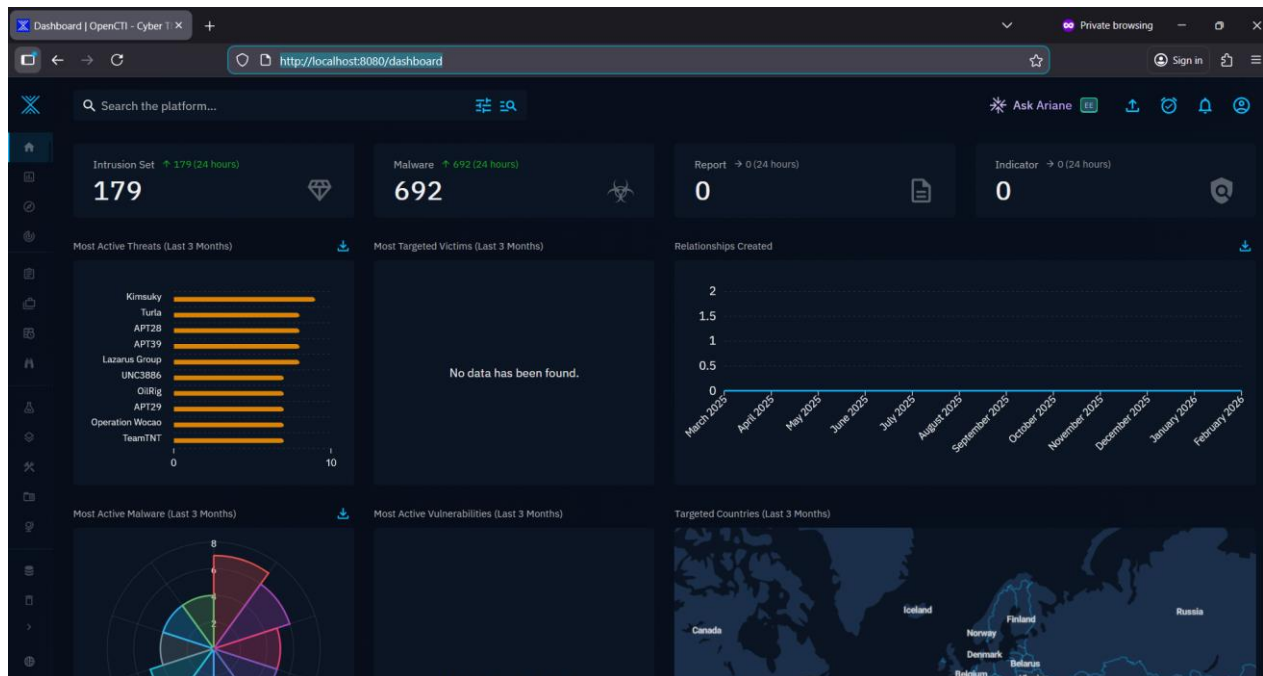
Issue 4: Login loop (redirect to login page)

- Symptoms: Successful login but redirected back
- Root Cause:
 - Admin password set as numeric (12345678)
 - Caused backend crypto/session failure
- Solution:
 - Changed password to:

Admin12345!

- Recreated containers
-

OpenCTI NOW IS WORKING AND I LOGGED IN SUCCEFULLY



Create Test Indicators

To validate integration, created sample Indicators of Compromise (IOCs):

IP Indicator

- Name: Malicious IP Test
- Pattern type: STIX
- Pattern:

[ipv4-addr:value = '1.2.3.4']

- Main observable: IPv4-Addr

The screenshot shows the OpenCTI Indicators page. The table lists the following indicators:

Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
Stix	Test URL Indicator	-	admin	-	Mar 18, 2026, 1:50:15 PM	-
Stix	Test Domain Indicator	-	admin	-	Mar 18, 2026, 1:49:47 PM	-
Stix	Malicious IP Test	-	admin	-	Mar 18, 2026, 1:47:22 PM	-

Domain Indicator

- Name: Test Domain Indicator
- Pattern type: STIX
- Pattern:

[domain-name:value = 'badexample.com']

- Main observable: Domain-Name

Observations / Indicators

Search these results... Add filter Select saved filter 1-3/3 Create Indicator

pattern type = and main observable type =

☐	Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
☐	Stix	Test URL Indicator	-	admin	-	Mar 18, 2026, 1:50:15 PM	- >
☐	Stix	Test Domain Indicator	-	admin	-	Mar 18, 2026, 1:49:47 PM	- >
☐	Stix	Malicious IP Test	-	admin	-	Mar 18, 2026, 1:47:22 PM	- >

URL Indicator

- Name: Test URL Indicator
- Pattern type: STIX
- Pattern:

[url:value = 'http://malicious-site.com']

- Main observable: URL

Observations / Indicators

Search these results... Add filter Select saved filter 1-3/3 Create Indicator

pattern type = and main observable type =

☐	Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
☐	Stix	Test URL Indicator	-	admin	-	Mar 18, 2026, 1:50:15 PM	- >
☐	Stix	Test Domain Indicator	-	admin	-	Mar 18, 2026, 1:49:47 PM	- >
☐	Stix	Malicious IP Test	-	admin	-	Mar 18, 2026, 1:47:22 PM	- >

Generate OpenCTI API Token

Navigated to:

Profile → API Tokens

Created new token:

Name: OTX

Copied token (used later in script) **I CREATED AND SAVED API TOKENS OF BOTH OTX AND OPEN CTI TO BE ABLE TO SEND DATA FROM OPEN CTI TO OTX**

Create OTX Account & API Key

- Logged into AlienVault OTX
 - Generated API Key from account settings
 - Saved API key for integration
 - **I CREATED AND SAVED API TOKENS OF BOTH OTX AND OPEN CTI TO BE ABLE TO SEND DATA FROM OPEN CTI TO OTX**
-

Install Python Dependencies

Opened Command Prompt and ran:

```
python -m pip install pycti requests
```

Create Integration Script

Created Python file:

```
notepad opencti_to_otx.py
```

Script functionality:

```

Python
from pycti import OpenCTIApiClient
import requests
import re

OPENCTI_URL = "http://localhost:8080"
OPENCTI_TOKEN = "PASTE_NEW_OPENCTI_TOKEN_HERE"
OTX_API_KEY = "PASTE_NEW_OTX_API_KEY_HERE"

api = OpenCTIApiClient(OPENCTI_URL, OPENCTI_TOKEN)

indicators = api.indicator.list(first=20)
print(f"Fetchd {len(indicators)} indicators from OpenCTI")

otx_indicators = []

for item in indicators:
    name = item.get("name")
    pattern = item.get("pattern", "")

    print(f"Checking: {name} | pattern: {pattern}")

    if not pattern:
        continue

    ip_match = re.search(r"\[ipv4-addr:value = '([^\']+)'\]", pattern)
    domain_match = re.search(r"\[domain-name:value = '([^\']+)'\]", pattern)
    url_match = re.search(r"\[url:value = '([^\']+)'\]", pattern)

    if ip_match:
        otx_indicators.append({
            "indicator": ip_match.group(1),
            "type": "IPv4"
        })
    elif domain_match:
        otx_indicators.append({
            "indicator": domain_match.group(1),
            "type": "domain"
        })
    elif url_match:
        otx_indicators.append({
            "indicator": url_match.group(1),
            "type": "URL"
        })

print("Prepared indicators for OTX:")
print(otx_indicators)

if not otx_indicators:
    print("No indicators found to send to OTX.")
    raise SystemExit

payload = {
    "name": "OpenCTI Export Test",
    "description": "Indicators exported from OpenCTI",
    "public": False,
    "indicators": otx_indicators
}

headers = {
    "X-OTX-API-KEY": OTX_API_KEY,
    "Content-Type": "application/json"
}

response = requests.post(
    "https://otx.alienvault.com/api/v1/pulses/create",
    json=payload,
    headers=headers
)

print("Status:", response.status_code)
print("Response:", response.text)

```

- Connect to OpenCTI using API token
- Retrieve indicators
- Extract IOC values from STIX patterns

- Send indicators to OTX as a pulse
-

Run the Script

I WROTE IN CMD:

```
python opencti_to_otx.py
```

Script Output

Observed:

Fetches 3 indicators from OpenCTI

Prepared indicators for OTX:

```
[{'indicator': 'badexample.com', 'type': 'domain'},  
 {'indicator': '1.2.3.4', 'type': 'IPv4'},  
 {'indicator': 'http://malicious-site.com', 'type': 'URL'}]
```

Status: 201 MEANS ITS WORKED

Verify in OTX WEBSITE

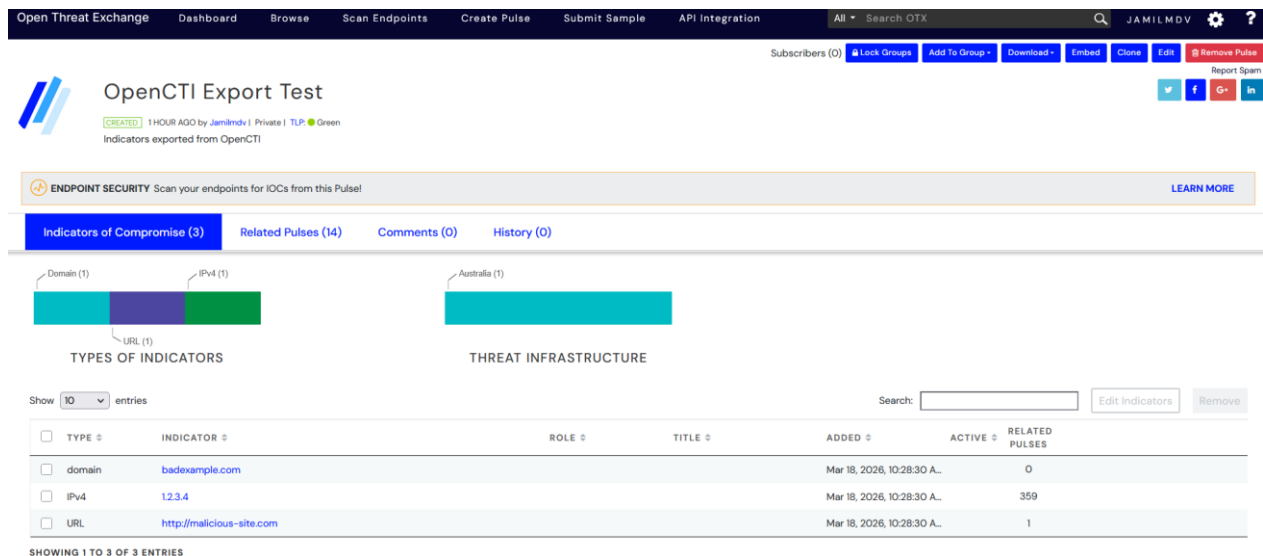
- Logged into AlienVault OTX
- Navigated to **My Pulses**
- Confirmed new pulse created:

OpenCTI Export Test

- Verified:

Indicators count: 3

Types: Domain, IPv4, URL



Second task

MalwareBazaar Integration with OpenCTI – Technical Report

Objective

The mission of this task was to integrate threat data like malware and hashes from MalwareBazaar to OpenCTI platform. I had to visualize malware related information in OpenCTI.

Environment Setup

OpenCTI Deployment

OpenCTI was deployed locally using Docker

Access was verified via:

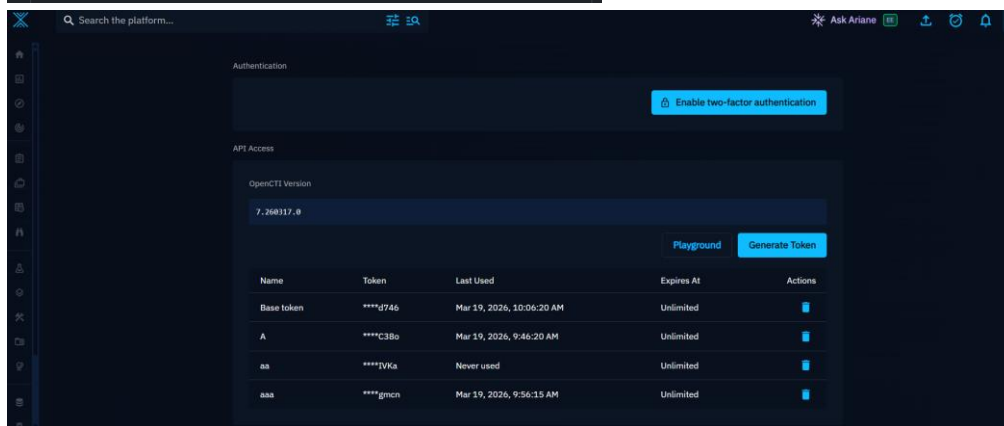
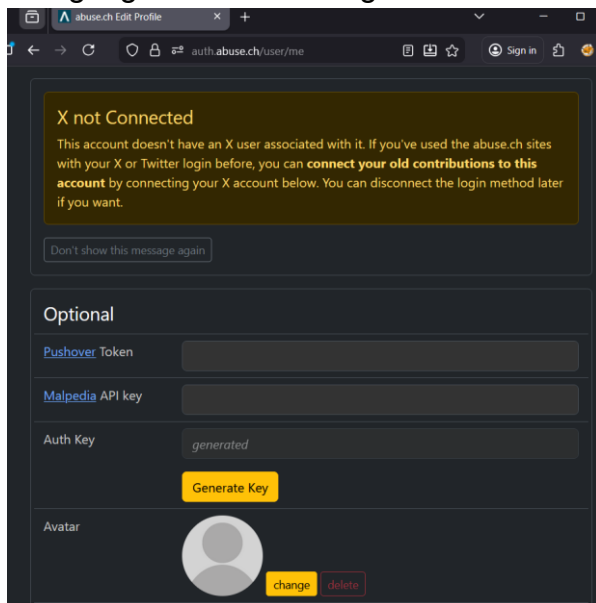
URL: <http://localhost:8080>

Admin credentials configured in .env (notepad through cmd)

API Tokens

OpenCTI API Token was generated from the platform settings. I need API tokens to be able to send my data from Malware Bazaar to OpenCTI by using python script in CMD

MalwareBazaar API Auth-Key was obtained for authenticated API requests. By doing login I was able to get API credentials from my profile in their website.



Initial Approach (Manual Import)

Attempt via File Import

I tried first to do it manually by just downloading daily report from Malware Bazaar and importing it to my account in OPENCTI, but it didn't work I could not see any data in my dashboard. Steps of process are these:

Accessed MalwareBazaar public dataset via:

<https://bazaar.abuse.ch/downloads/misp/>

Downloaded JSON files (MISP format).

Attempted to import them into OpenCTI using:

- OpenCTI → Import → Upload JSON file

Issues Encountered

- Browser displayed JSON instead of downloading (resolved using **Ctrl + S**).
 - Some files were incorrectly saved with .txt extension.
 - Import process did not consistently reflect visible data in the dashboard.
-

API-Based Integration (Final Approach)

Due to limitations of manual import, a programmatic approach using Python was implemented. This was more effective because I used API Credentials here

Script Purpose

The script performs:

1. Fetch recent malware samples from MalwareBazaar API (My account)
 2. Extract relevant fields (signature, tags, file name)
 3. Send data to OpenCTI
 4. Create malware objects inside OpenCTI
-

Implementation Details

MalwareBazaar API Usage

I opened cmd and first created notepad to send my data from bazaar to opencti

```
malwarebazaar_to_openciti.py malware_only_to_openciti.py
```

```
File Edit View
```

```
from pycti import OpenCTIApiClient
import requests

OPENCTI_URL = "http://localhost:8080"
OPENCTI_TOKEN = "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXZWQifQ.eyJ1bm9udGVudCI6ImFkbWVudCJ9"
MBAZAAAR_AUTH_KEY = "MBAAZAAAR-Auth-Key"

api = OpenCTIApiClient(OPENCTI_URL, OPENCTI_TOKEN)

headers = {
    "Auth-Key": MBAZAAAR_AUTH_KEY
}

data = {
    "query": "get_recent",
    "selector": "100"
}

response = requests.post(
    "https://mb-api.abuse.ch/api/v1/",
    headers=headers,
    data=data,
    timeout=60
)

print("MalwareBazaar HTTP status:", response.status_code)
```

```
Created malware: AsyncRAT
INFO:api:Creating Malware
Created malware: SalatStealer
INFO:api:Creating Malware
Created malware: RemcosRAT
INFO:api:Creating Malware
Created malware: Phorpiex
INFO:api:Creating Malware
Created malware: NetWire
INFO:api:Creating Malware
Created malware: njrat
INFO:api:Creating Malware
Created malware: OffLoader
INFO:api:Creating Malware
Created malware: GuLoader
INFO:api:Creating Malware
Created malware: Cobalt Strike
INFO:api:Creating Malware
Created malware: Mirai
INFO:api:Creating Malware
Created malware: RedLineStealer
INFO:api:Creating Malware
Created malware: DarkTortilla
INFO:api:Creating Malware
Created malware: Gh0strAT
INFO:api:Creating Malware
Created malware: EternalRocks
INFO:api:Creating Malware
Created malware: LummaStealer
INFO:api:Creating Malware
Created malware: Efimer
INFO:api:Creating Malware
Created malware: Vidar
Done. Malware objects created: 19
PS C:\Users\Jamil> notepad malware_only_to_openciti.py
PS C:\Users\Jamil> |
```

MalwareBazaar IMPORT FILE Error

Received HTML instead of JSON when using incorrect request type. Here I downloaded a wrong File, when I first trying to download

Resolved by:

Downloading a correct file with json, but still I could not see any visible data.

OpenCTI Visibility Issue

Data not immediately visible on dashboard

I understood that I need to go to:

Entities → Malware

Observations → Indicators

Results

Successful Outcomes

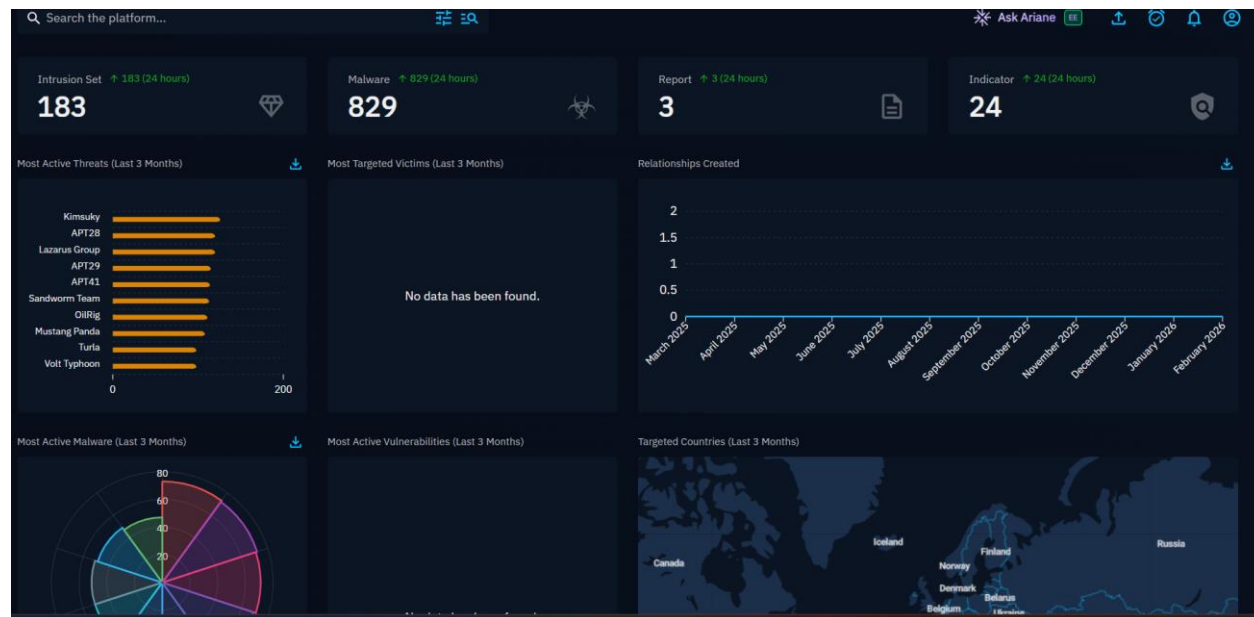
Successfully retrieved malware data from MalwareBazaar API

Created multiple **Malware entities** in OpenCTI such as:

njrat

Mirai

RemcosRAT



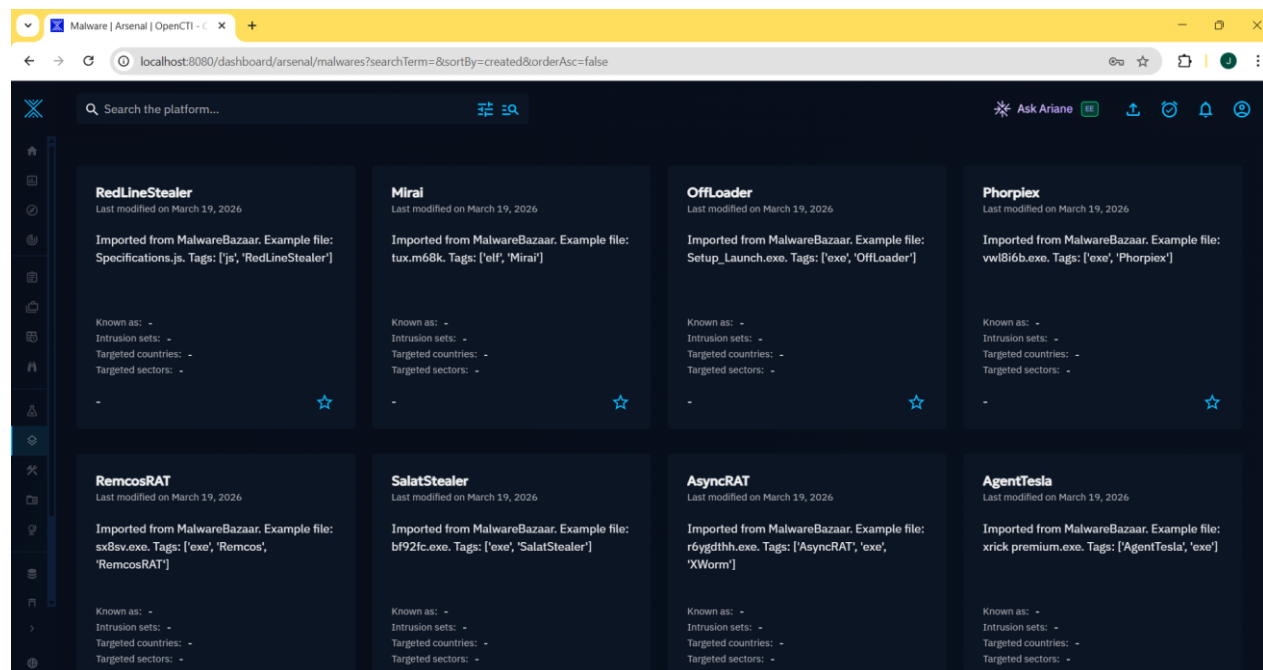
Platform Verification

Data confirmed visible in:

Entities → Malware

Indicators (from previous steps) visible under:

Observations → Indicators



Conclusion

The integration of MalwareBazaar with OpenCTI was successfully achieved using a Python-based API approach. This method was more reliable than manual file import and allowed automated ingestion of threat intelligence data.

Task 3

AbuseIPDB to OpenCTI Integration Report

Objective

The purpose of this task was to integrate **AbuseIPDB** with **OpenCTI** to import abusive IP address in OpenCTI and make it visible as indicators and reports (from indicators it had to be linked in reports) inside the platform.

Goal of the Integration

I had to retrieve abusive IP addresses from AbuseIPDB through its API, then transform the returned data into OpenCTI-compatible objects, after which create IP indicators in OpenCTI and group the imported indicators under a report for easier visibility and tracking

Data Source

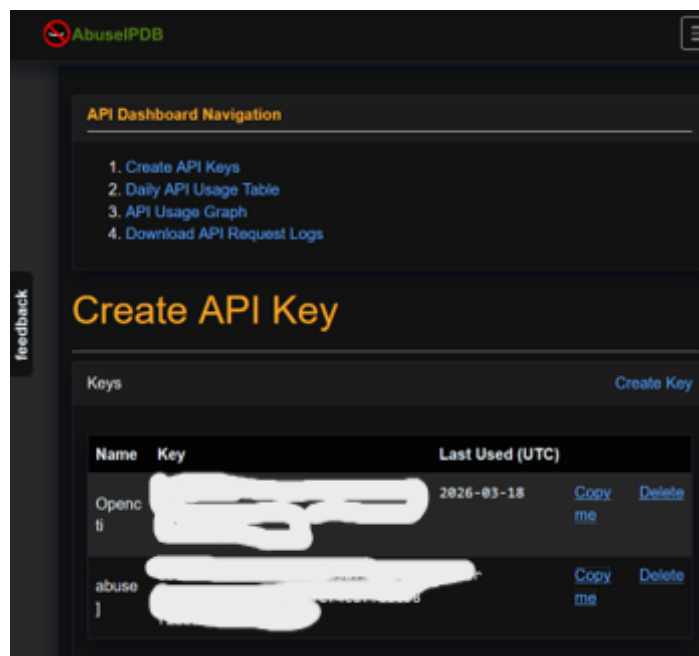
I used data from the **AbuseIPDB blacklist API**, which provides a list of abusive IP addresses based on abuse confidence score..

1 Step

Two tokens/keys were required:

- **OpenCTI API token**
Used to authenticate against the OpenCTI API and create objects inside the platform.
- **AbuseIPDB API key**
Used to authenticate requests to the AbuseIPDB API.

So for that from both of them I entered to profile to get their API credentials.



Step 2 – Connect to AbuseIPDB

The script sent an authenticated request to the AbuseIPDB blacklist API. I created script in chatgpt for requesting data from AbuseIPDB blacklist API then I added that script to notepad

```
C:\Users\Jamil>notepad abuseipdb_to_opencti.py
```

```
C:\Users\Jamil>python abuseipdb_to_opencti.py
```

Step 3 – Retrieve IP Data

The API returned a large list of abusive IP addresses, including metadata such as:

IP address

abuse confidence score

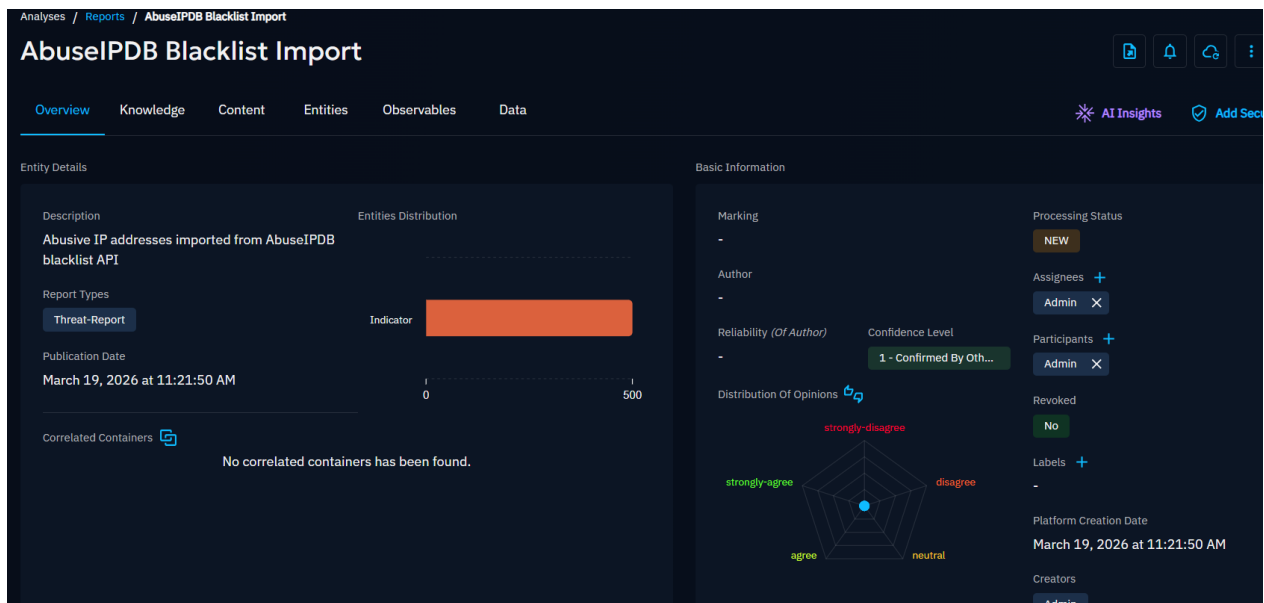
usage type

associated domain

```
INFO:api:Creating Indicator
Created indicator: 68.178.161.186
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 85.203.23.125
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 95.210.105.166
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 103.171.69.112
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 47.250.184.124
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 80.94.92.67
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 170.64.183.150
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 125.16.27.190
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 31.173.8.170
INFO:api:Adding StixObjectOrStixRelationship to Report
INFO:api:Creating Indicator
Created indicator: 204.76.203.56
```

Step 4 – Create Report in OpenCTI

A report object was created in OpenCTI to act as a container for all imported indicators.



Step 5 – Create Indicators

Each abusive IP was transformed into an OpenCTI indicator using STIX format:

```
[ipv4-addr:value = 'IP_ADDRESS']
```

Main observable type:

IPv4-Addr

Observations / Indicators

Search these results...

Add filter

Select saved filter

1 - 25 / 542

pattern type =

and

main observable type =

☐	Pattern type	Name	Author	Creators	Labels	Original creation date	Marking
☐	Stix	AbuseIPDB IP 41.90.100.147	-	admin	-	Mar 19, 2026, 11:23:21 AM	-
☐	Stix	AbuseIPDB IP 39.97.253.143	-	admin	-	Mar 19, 2026, 11:23:21 AM	-
☐	Stix	AbuseIPDB IP 154.209.4.208	-	admin	-	Mar 19, 2026, 11:23:20 AM	-
☐	Stix	AbuseIPDB IP 116.140.211.29	-	admin	-	Mar 19, 2026, 11:23:20 AM	-
☐	Stix	AbuseIPDB IP 34.72.176.129	-	admin	-	Mar 19, 2026, 11:23:20 AM	-
☐	Stix	AbuseIPDB IP 152.32.144.167	-	admin	-	Mar 19, 2026, 11:23:20 AM	-
☐	Stix	AbuseIPDB IP 20.168.1.102	-	admin	-	Mar 19, 2026, 11:23:20 AM	-
☐	Stix	AbuseIPDB IP 87.248.237.138	-	admin	-	Mar 19, 2026, 11:23:19 AM	-
☐	Stix	AbuseIPDB IP 103.187.147.165	-	admin	-	Mar 19, 2026, 11:23:19 AM	-

Step 6 – Link Indicators to Report

Each created indicator was linked to the report so that the report page in OpenCTI would show imported entities. Special script was created to be able to connect them

Initial Issues Encountered

Report Created but No Data Visible

At first, the report appeared empty even though indicators were created successfully.

Cause:

Indicators were created in OpenCTI but were not linked to the report.

Resolution:

The script was updated to add each indicator to the report individually using the correct report-linking method. I changed it with the help of chatgpt , I saw an error in notepad and created a new script

Data Imported

The import included:

- IP addresses from AbuseIPDB blacklist

- abuse confidence score

- country code

- ISP

- usage type

- domain

In one successful run:

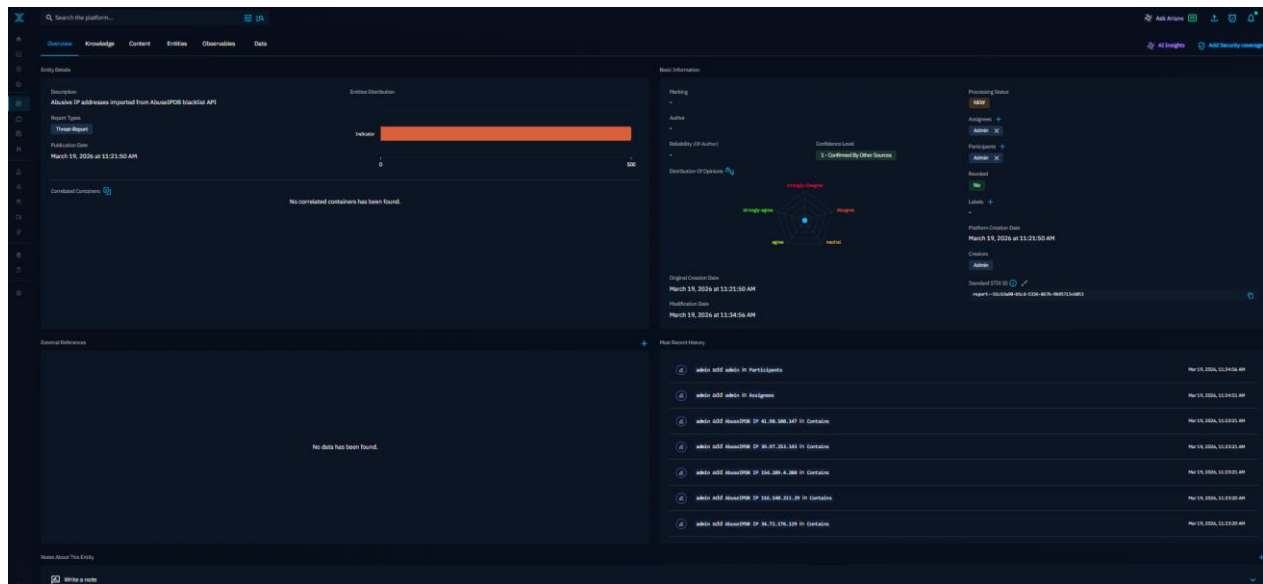
- 500 indicators** were created and linked to the report (First it was sending just 20 indicators, but I changed the number of it to 500 in script)

Verification in OpenCTI

Report Verification

The report page in OpenCTI showed an **Entities Distribution** section containing indicators.

This confirmed that the indicators were successfully linked to the report.



Conclusion

The integration between AbuseIPDB and OpenCTI was successfully implemented.